

HOW ELECTRONIC NOSE Is Better Than Ours



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An electronic nose, or e-nose, is a device that identifies specific components of an odour and analyses its chemical makeup to identify it. It is engineered to mimic the mammalian olfactory system but is better in some respects.

How it works

Air sample is pulled by a vacuum pump and led through a tube into a small chamber consisting of an electronic sensor array. A transient response is produced as the volatile organic compounds in the sample interact with the surface of the sensor's active material. A steady-state response is reached within a few minutes, which is then sent to a signal-processing unit.

A washing gas, like an alcohol vapour, is applied to the array for approximately one minute to remove the

odourant mixture from the surface and bulk of the sensor's active material.

At last, reference gas is applied to the array to prepare it for a new measurement cycle. The pattern of responses across all sensors in the array is used to identify the odour.

An e-nose includes three major parts:

1. Sample delivery system
2. Detection system
3. Computing system

Sample delivery system. This enables the generation of headspace (volatile compounds) for a sample. The system then injects it into the detection system of the e-nose.

Detection system. A sensor set is the reactive part of the system. Adsorption of volatile compounds on the surface of the sensor causes a physical change in the sensor, where it experiences a change in electrical properties. A specific response is recorded by the electronic interface, transforming the signal into a digital value. Recorded data is then computed based on statistical models.

Computing system. This system works to combine the responses of all sensors.

Sensors

A sensor array is the key element of an e-nose. In most e-noses, sensors are sensitive to volatile molecules but in different ways. However, in bio-e-noses, receptor proteins that respond to specific odour molecules are used.

The most commonly-used sensors are:

Conductivity sensors. These can be classified as follows:

Metal oxide sensors (MOsEs).

Adsorption of gas molecules provokes changes in conductivity. This conductivity change is the measure of total volatile organic compounds adsorbed.

